# Concept-Level Unlearning for Vision-Language Models

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## Abstract

Vision-language models (VLMs) are increasingly used in applications where the ability to erase recognition of entire visual concepts (rather than individual images) is required for privacy, safety or regulatory compliance. Existing multimodal unlearning methods, however, predominantly target sample- or instance-level forgetting, leaving image-conditioned concept-level unlearning largely unaddressed. We propose CMU (Concept-level Machine Unlearning), an adapter-only method that reduces image-conditioned target-class recognition while preserving non-target visual recognition and general language ability. CMU freezes the vision encoder and LLM, trains only a lightweight adapter, selects the most suitable semantic LLM layer by a concept-separability criterion, and applies all representation supervision exclusively at that layer. Its forgetting loss uses adaptive anchor alignment to match target-class representations to a fixed neutral distribution, while retention and regularization preserve original capabilities. Experiments on LLaVA, Qwen2.5-VL and Qwen3-VL across ImageNet-100 and the Risky Visual Concepts benchmark show that CMU achieves superior and more consistent performance in forgetting and retention than existing baselines across models and datasets.

## Introduction

Modern generative vision-language models (VLMs), such as LLaVA and Qwen-VL, achieve strong visual understanding and multimodal reasoning (Liu et al. 2023; Bai et al. 2025b). However, knowledge acquired from large-scale image-text corpora may later require removal for privacy, safety, or data-protection reasons, including erasure rights under the GDPR (Liu et al. 2025a; Chen et al. 2025a; European Parliament and Council of the European Union 2016). Machine unlearning enables targeted removal without costly full retraining (Cheng and Amiri 2024; Huo et al. 2025).

During large-scale pre-training, VLMs learn from massive image-text corpora and encode a broad spectrum of knowledge within their parameters. However, some of this knowledge may introduce practical risks and therefore motivate targeted intervention. First, privacy-sensitive identities and personal information may be embedded in models, conflicting with data-protection regulations and the right to be forgotten. Second, copyrighted works, artistic styles, and protected visual entities may raise intellectual-property concerns. Third, unsafe or undesirable content may lead to harmful or policy-violating behaviors. These concerns motivate **concept-level unlearning for VLMs**. Here, a *concept* is a category's shared visual semantics, while a *class* is its dataset-level operationalization; target-class images instantiate the target concept.

Existing VLM-unlearning settings often center on particular image-text pairs, images, or entities (Cheng and Amiri 2024; Li et al. 2024a; Huo et al. 2025). Naively extending sample-wise deletion to a whole concept requires covering many related instances, while stronger updates can impair retained multimodal capabilities (Huo et al. 2025; Liu et al. 2025b).

Although concept-level unlearning has been explored for CLIP and diffusion models, these methods target representations or image generation rather than VLM understanding and do not transfer directly.

Image-conditioned concept-level unlearning in VLMs raises three key issues. From the perspective of generalization, a concept can appear across diverse visual and linguistic patterns, making it difficult to reduce recognition beyond the observed training samples. From the perspective of precision, the intervention should reduce target recognition while limiting changes to unrelated concepts and general multimodal capabilities. From the perspective of efficiency, the scale of VLMs makes it important to identify a small, relevant parameter subset for optimization.

To address these challenges, we propose **CMU** (Concept-level Machine Unlearning), a framework that realigns internal representations in adapter-based VLMs. Rather than updating the entire model, CMU freezes the backbone and only updates the adapter during unlearning, while applying all forgetting supervision at a single informative layer along the visual-to-language pathway. Forget-class representations are matched as a set to a fixed neutral reference distribution, retain-class representations are anchored to their pretrained states, and adapter drift is explicitly regularized. This provides an adapter-focused intervention for reducing image-conditioned target recognition while constraining changes to non-target representations. Our contributions are as follows.

- **CMU Framework:** To our knowledge, CMU is the first framework for visual concept-level unlearning in VLMs. It combines adapter-only optimization with supervision at a selected LLM layer to reduce target-concept recognition while preserving non-target visual recognition and general language capabilities.

- **Adapter-only Unlearning:** We confine unlearning to the lightweight adapter between the vision encoder and LLM while keeping both backbone modules frozen. This design intervenes directly in visual information transfer, substantially reduces the number of trainable parameters, preserves general language capabilities, and remains robust to the textual prompt attacks evaluated in our experiments.
- **Layer-Wise Supervision:** We develop a data-dependent strategy that identifies a single class-discriminative LLM layer and applies all representation supervision exclusively at that layer. It aligns forget representations with a neutral reference distribution while anchoring retain representations to their pretrained states, thereby concentrating the unlearning signal and constraining changes to non-target representations.
- **Extensive Experiments:** We conduct extensive evaluations on ImageNet-100 and our 47-class RVC benchmark across three VLMs. The results demonstrate consistently strong forgetting and capability preservation across different models and datasets.

## Related Work

Recent work on machine unlearning in MLLMs has primarily focused on **instance-level or identity-level erasure**, such as removing specific facial identities (Huo et al. 2025; Bhaila et al. 2025), personal profiles (Zeng et al. 2025), or individual image-text pairs (Cheng and Amiri 2024; Li et al. 2024a). Early methods (Cheng and Amiri 2024; Chakraborty et al. 2024) extended unimodal techniques to the multimodal setting, but found that forgetting in one modality often leaves sensitive information recoverable through cross-modal pathways.

Subsequent works improved precision via weight-saliency-based selection (Huo et al. 2025), modality-aware pruning (Liu et al. 2025b), and anchor-guided masking (Zeng et al. 2025; Li et al. 2024a). However, these approaches target fine-grained instances or identities, whereas we ask whether an adapter-based VLM can unlearn an entire visual category. In parallel, harmful-content unlearning (Chakraborty et al. 2024; Chen et al. 2025b) is typically framed as alignment or refusal learning rather than visual class recognition removal.

Concept-level unlearning has also been studied in more restricted vision-language settings, especially CLIP-style representation models (Kravets and Namboodiri 2025a; Yang et al. 2025; Mishra et al. 2025; Kravets and Namboodiri 2025b). These methods provide useful evidence that visual concepts can be weakened at the category level without retraining a full model. However, CLIP unlearning does not directly address generative VLMs whose visual tokens are translated into an LLM and then used for free-form answers. CMU targets this adapter-based generative setting by modifying only the adapter that transfers visual evidence into the frozen language model.

From a methodological perspective, the dominant paradigm remains **gradient ascent (GA) and its variants** (Chakraborty et al. 2024; Maini et al. 2025; Zhang et al. 2024; Huo et al. 2025; Liu et al. 2025b; Cai et al. 2026). Despite increasingly sophisticated constraints, these methods inherit GA's well-known weaknesses: sensitivity to hyperparameters, retain-set degradation, and vulnerability to adversarial recovery attacks that expose the shallow nature of loss-surface-level forgetting (Zhang et al. 2025). A closely related representation-level method is Representation Misdirection for Unlearning (RMU) (Li et al. 2024b), which steers forget representations toward a random direction while preserving non-target behavior. CMU shares the idea that representation-level supervision can be more direct than output-only GA, but differs in three ways: it operates on image-conditioned visual-to-language transfer in VLMs, restricts updates to the adapter, and replaces random misdirection with a single-layer adaptive anchor that aligns representations to a neutral distribution estimated from non-target visual references. Our anchor-target ablation directly probes this design choice.

Beyond these weaknesses, existing methods treat modalities as separable via pruning (Liu et al. 2025b), attention perturbation (Bhaila et al. 2025), or decoupled losses (Cheng and Amiri 2024). While effective for instance-level privacy removal, they cannot control how visual category evidence shapes language-side reasoning, so image-driven concept-level unlearning in generative VLMs remains unresolved.

## Motivation

Effective unlearning is essential for removing unwanted knowledge while preserving a model's overall capabilities. Achieving efficient concept-level unlearning with minimal disruption mainly depends on two aspects of model parameter updates: selecting the most effective model component and designing an appropriate supervision signal. The first facilitates targeted forgetting, while the second helps preserve the model's retained capabilities.

This raises two key questions in designing CMU:

- How can we define a parameter-efficient subset of tunable parameters for concept-level unlearning in VLMs?
- How can we design effective supervision signals to guide adapter-only unlearning in VLMs?

For the first question, we observe that a typical VLM architecture consists of three sequential components: a vision encoder (e.g., ViT), an adapter module, and a large language model (LLM). Among these components, both the vision encoder and the LLM are well-established pretrained modules with a large number of parameters. In contrast, the adapter is a lightweight module that maps visual representations into the language space, thereby connecting the visual and language modalities.

Based on this observation, we perform unlearning by updating only the adapter. On the one hand, it plays a critical role in aligning visual representations with the language space, making it highly relevant for controlling class-specific visual knowledge. On the other hand, it contains significantly fewer

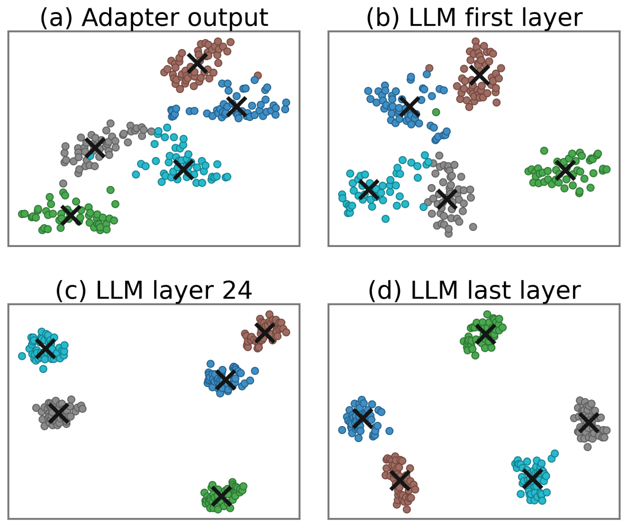


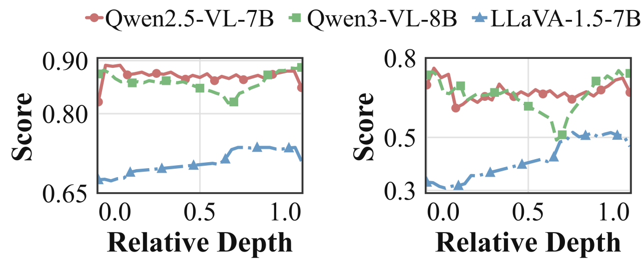

Figure 1: t-SNE visualization of per-class representations at representative positions in LLaVA-1.5. Class separability varies across depth, with an intermediate semantic layer showing the clearest clustering.

parameters than the full model, enabling more efficient unlearning while reducing the risk of degrading the model's general linguistic capabilities. That is, we freeze the ViT and LLM backbones and only update the adapter in CMU.

For the second question, inspired by the distortion of target output for unlearning in (Seo, Kim, and Han 2025; Mirlach, Laguna, and Vogt 2026; Wang et al. 2024; Vasilev et al. 2025), a natural choice is to distort the output of the adapter updated during unlearning, as it directly reflects the updated module. However, VLMs provide additional supervision sources, particularly from the downstream LLM with strong representational capacity, raising the question of which layer output to distort.

We argue that effective unlearning requires supervision from layers with strong class discriminability, as such layers provide better class separation, facilitating the removal of target-class representations while minimizing interference with other classes. Moreover, we observe that class-specific structure is progressively transformed through the frozen LLM, becoming more explicitly separable at certain depths as the adapter outputs propagate across layers. This intuition is supported by empirical analysis in Figure 1, which illustrates the evolving clustering behavior of features across layers, while Figure 2 shows the corresponding variation in layer separability. Based on these observations, we propose to distort the output of the layer with the strongest classification capability, as determined by statistical criteria, to guide forgetting.

The above analysis motivates our CMU design, where we perform adapter-only unlearning using supervision signals derived from distorting the output of the LLM layer with the strongest class discriminability.




(a) KNN accuracy (b) Silhouette score

Figure 2: Layer separability across normalized depth; the non-flat profiles motivate data-dependent layer selection.

| Symbol | Description |
|---|---|
| $X_F$, $X_R$, $X_N$ | Forget, retain, and non-target reference sets |
| $\mathcal{E}$, $\mathcal{A}$, $f_{\text{LM}}$; $\theta_A$ | Encoder, adapter, LLM, and trainable adapter parameters |
| $H^{(k)}$, $z^{(k)}$, $\widehat{z}^{(k)}$ | Hidden, pooled, and normalized representations |
| $\mathcal{K}$, $k^\star$; $A^{(k)}$, $S^{(k)}$ | Candidate/selected layers and calibration scores |
| $\mathcal{B}$; $\kappa$, $\Sigma$ | Reference bank, RBF kernel, and bandwidths |
| $\gamma$, $\lambda$, $\mu$ | Forget, retain, and parameter-loss weights |

Table 1: Key notation.

# Methodology

## Problem Formulation

We consider concept-level image-driven unlearning for an adapter-based VLM: image $\to \mathcal{E} \to \mathcal{A} \to f_{\text{LM}}$. Throughout CMU, $\mathcal{E}$ and $f_{\text{LM}}$ stay frozen; only $\mathcal{A}$ is trainable. Given forget set $X_F$ (target class) and retain set $X_R$ (non-target classes), the goal is to find updated adapter parameters that suppress target-class recognition from visual inputs while preserving performance on non-target data. This formulation intentionally targets the visual-to-language pathway rather than all knowledge stored in the frozen backbones.

For image $x$, visual tokens after the adapter are $H^{(0)}(x) = \mathcal{A}(\mathcal{E}(x))$; for $k \geq 1$, $H^{(k)}(x)$ is the hidden state tensor at LLM layer $k$. Every calibration, bank-construction, forget, and retain pass uses the same fixed empty text prompt. We define $z^{(k)}(x) \in \mathbb{R}^d$ by mean-pooling the hidden vectors of all *visual* token positions at layer $k$ (positions corresponding to the image stream; no averaging over text tokens), where $d$ is the hidden dimension, and write $\widehat{z}^{(k)}(x) = z^{(k)}(x)/\|z^{(k)}(x)\|_2$ for its row-wise normalized form. CMU supervises only one selected LLM layer $k^\star$. The candidate layers are $\mathcal{K} = \{1, \dots, L\}$, where $L$ is the total number of LLM layers, or a contiguous subset thereof depending on the backbone. We denote the adapter weights at the start of unlearning as $\theta_A^{\text{init}}$.

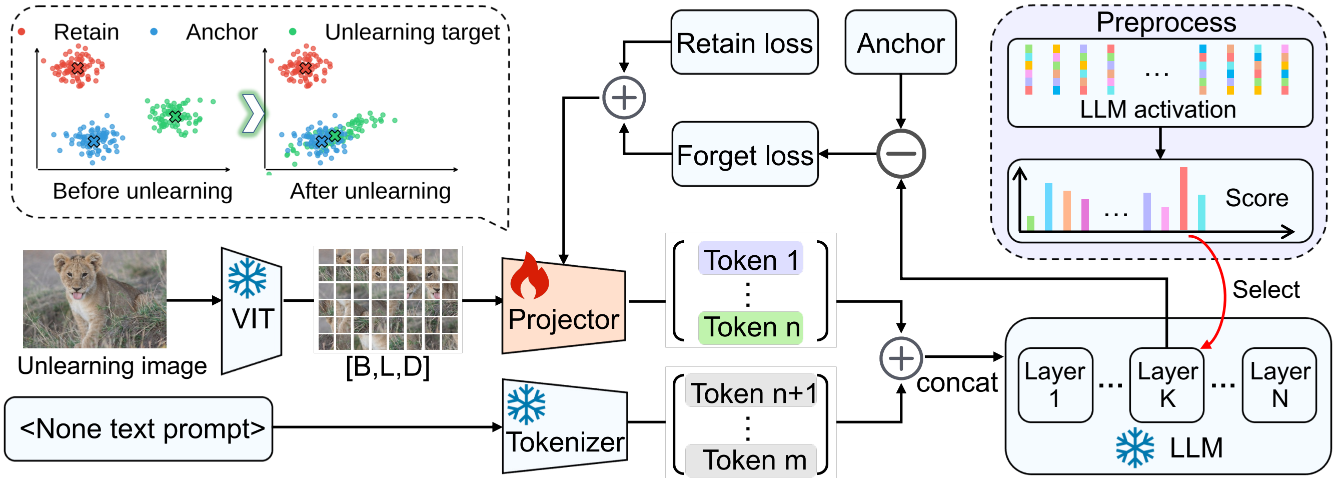



Figure 3: Overview of CMU. Visual features from the frozen vision encoder pass through a trainable adapter to a frozen LLM. CMU combines data-dependent layer selection, single-layer adaptive-anchor alignment, and adapter-only unlearning.

## Overview

CMU addresses concept-level unlearning in adapter-based VLMs by confining all parameter updates exclusively to the vision-language adapter $\mathcal{A}$, while keeping both the vision encoder and the LLM frozen. It leverages adaptive anchors and precise concept-level supervision to relocate forget-set representations away from their original conceptual regions in the embedding space. The method comprises three coordinated components, illustrated in Figure 3, with key notation summarized in Table 1; detailed descriptions follow below.

- **Layer-wise supervision.** During unlearning, CMU identifies a single semantic LLM layer $k^\star$ by evaluating each candidate layer based on how distinctly its pooled visual token representations separate class identities. No loss is applied to the adapter output or to any other LLM layer.
- **Adaptive anchor alignment.** CMU constructs a class-balanced set of non-target reference representations from an image-disjoint reference set at the selected semantic layer. The forget representations are aligned with these references as a whole, rather than being paired with individual samples or collapsed toward one artificial target. The reference representations are computed before optimization and remain fixed during training.
- **Adapter-only Unlearning.** Only the vision-language adapter $\mathcal{A}$ is updated during unlearning. Gradients backpropagate through the frozen LLM to reach $\mathcal{A}$, leaving all backbone weights unchanged.

## Layer-Wise Supervision

Because class structure varies across depth (Figure 1), CMU selects exactly one semantic layer $k^\star$ using a labeled visual calibration set of $n$ classes with $m$ images per class. We select $k^\star$ once before unlearning for each backbone and hold it fixed across that backbone's runs. The calibration set serves only to compare candidate-layer class separation.

For each candidate layer $k \in \mathcal{K}$ and each calibration image $x$, we extract token-level hidden states $H^{(k)}(x)$ using the initial adapter weights and mean-pool the visual-token positions as defined in the Problem Formulation subsection to obtain $z^{(k)}(x)$.

To score layer $k$, let $\{\mathbf{z}_i\}_{i=1}^{nm}$ denote the pooled representations $z^{(k)}(x_i)$ of all calibration images. Using one fixed stratified split for every candidate layer, we fit a linear probe on the calibration-training representations and record its held-out accuracy $A^{(k)}$. We also compute a true-label silhouette on the unnormalized pooled representations using Euclidean distance: for image $i$, $a_i^{(k)}$ is its mean distance to the other calibration images with the same class label, and $b_i^{(k)}$ is the smallest mean distance to any other class. Thus,

$$s_i^{(k)} = \frac{b_i^{(k)} - a_i^{(k)}}{\max\{a_i^{(k)}, b_i^{(k)}\}}, \qquad S^{(k)} = \frac{1}{nm}\sum_{i=1}^{nm} s_i^{(k)},$$

where larger $S^{(k)}$ indicates more compact and better-separated class-conditional neighborhoods. We rank layers lexicographically by held-out probe accuracy and then silhouette,

$$k^\star = \operatorname*{lex\,argmax}_{k\in\mathcal{K}}\big(A^{(k)}, S^{(k)}\big),$$

using $S^{(k)}$ only to break probe-accuracy ties. This $k^\star$ is the only layer used by the forget and retain representation losses. The split, probe optimizer, and seed are held fixed across layers; $k$-nearest-neighbor accuracy is reported only as a complementary diagnostic.

## Adaptive Anchor Alignment

The non-target reference set $X_N$ is a fixed collection of images used only to construct a reference bank at $k^\star$. We require $X_N$ to be image-disjoint from $X_F$ and from the retain samples used for training or evaluation, and we exclude all forget-class images from $X_N$. When a dataset provides auxiliary non-evaluation classes, $X_N$ can also be class-disjoint from both $X_F$ and $X_R$; when the benchmark partition exhausts the available class labels, the essential requirement

**Evaluation Prompts**

*$C$ denotes the ground-truth class; $\{d_i\}$ are distractors sampled from non-target classes.*

**I: MCQ.** *Q.* What is the main object in this image?

*A.* $d_1$ *B.* $C$ *C.* $d_2$ *D.* $d_3$
*E.* None of the options

*Response.* One uppercase letter: A, B, C, D, or E.

**II: YN.** *Q.* Does the main object in this image belong to class $C$?
*Response.* One word: Yes or No.

**III: Open-Gen.** *Q.* What is the main object in this image? Describe it in detail.
*Response.* A short descriptive English phrase or sentence.

Figure 4: Evaluation prompt templates for the three question families. Appendix A specifies exact parsing, invalid-output, and class-aggregation rules.

is image-level disjointness and exclusion of the forget class. Images in $X_N$ are drawn from the same broad visual domain when possible, are not scored as retain data, and are held fixed across the unlearning run.

We select $M$ reference images from $X_N$ with a fixed class-balanced procedure and a per-class cap, and denote the selected subset by $X_N^{\text{ref}}$. We compute their selected-layer representations under the initial adapter. After row-wise $\ell_2$ normalization, the resulting fixed reference bank is

$$\mathcal{B} = \left\{ \widehat{z}^{(k^\star)}(x_n; \theta_A^{\text{init}}) \right\}_{x_n \in X_N^{\text{ref}}}, \qquad |\mathcal{B}| = M.$$

The bank is stored after layer selection and is never updated during adapter optimization. It preserves a diverse set of non-target reference features instead of reducing the target to a single center.

With $k^\star$ and $\mathcal{B}$ fixed, we train only $\mathcal{A}$. Let $\widehat{Z}_F = \{\widehat{z}_i\}_{i=1}^N$ be the row-wise $\ell_2$-normalized selected-layer representations of a forget minibatch, and write $\mathcal{B} = \{b_a\}_{a=1}^M$. We use a uniform mixture of RBF kernels,

$$\kappa(u, v) = \frac{1}{|\Sigma|} \sum_{\sigma \in \Sigma} \exp\left( -\frac{\|u - v\|_2^2}{2\sigma^2} \right),$$

to define the adaptive anchor alignment discrepancy

$$\begin{aligned} &\mathcal{D}_\kappa(\widehat{Z}_F, \mathcal{B}) \\ &= \frac{1}{N^2} \sum_{i,i'} \kappa(\widehat{z}_i, \widehat{z}_{i'}) + \frac{1}{M^2} \sum_{a,a'} \kappa(b_a, b_{a'}) \\ &\quad - \frac{2}{NM} \sum_{i,a} \kappa(\widehat{z}_i, b_a), \end{aligned}$$

where $N = |\widehat{Z}_F|$. The forget objective is

$$\mathcal{L}_{\text{forget}} = \gamma\, \mathcal{D}_\kappa(\widehat{Z}_F, \mathcal{B}),$$

with a fixed calibration scale $\gamma$. Minimizing this discrepancy aligns the forget representations with the diverse non-target reference bank at the single selected layer without relocating them toward a named competing class. The comparison is made between the two sets as a whole, rather than by pairing individual forget images with individual reference images. This differs from output-logit gradient ascent and from pointwise representation misdirection.

## Adapter-only Unlearning

Let $B_R \subseteq X_R$ denote a retain minibatch. Unlike the normalized representations used by the adaptive anchor alignment objective, the retain term operates on the unnormalized pooled representations $z^{(k^\star)}$. It averages the representation distance over images $x_r \in B_R$, anchoring both the direction and magnitude of each retain representation to its value under the initial weights:

$$\mathcal{L}_{\text{retain}} = \frac{1}{|B_R|} \sum_{x_r \in B_R} \left\| z^{(k^\star)}(x_r; \theta_A) - z^{(k^\star)}(x_r; \theta_A^{\text{init}}) \right\|_2.$$

The regularization term bounds the overall magnitude of adapter change using the Euclidean $\ell_2$ distance over the vectorized adapter parameters:

$$\mathcal{L}_{\text{param}} = \left\| \text{vec}(\theta_A) - \text{vec}(\theta_A^{\text{init}}) \right\|_2.$$

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{forget}} + \lambda\, \mathcal{L}_{\text{retain}} + \mu\, \mathcal{L}_{\text{param}},$$

where $\lambda$ and $\mu$ trade off the retain and parameter terms. Although $\mathcal{L}$ is evaluated at $k^\star$, autodiff backpropagates through the frozen lower LLM blocks to $\mathcal{A}$. Neither the encoder nor the LLM receives an optimizer step; only $\theta_A$ is updated. Appendix F specifies layer calibration, reference-bank sampling, RBF bandwidths, and hyperparameter selection. We select the optimizer, learning rate, and loss weights $\gamma$, $\lambda$, and $\mu$ through small-sample pilot experiments on non-forget datasets, and fix the selected model-level setting before the reported unlearning evaluations.

# Experiments

## Experimental Setup

We test CMU on three VLMs: LLaVA-1.5-7B (Liu et al. 2023), Qwen2.5-VL-7B (Bai et al. 2025b), and Qwen3-VL-8B (Bai et al. 2025a). Their official pretrained checkpoints serve as the Vanilla references. Across all CMU runs, only the vision-language adapter is updated; the vision encoder and LLM remain frozen.

**Datasets.** We evaluate on two visual forget benchmarks: ImageNet-100, a representative 100-class subset of ImageNet (Deng et al. 2009), and Risky Visual Concepts (RVC) v1, our 47-class image-backed safety concept benchmark. RVC is assembled from HAZMAT13, HOD, Dangerous Items 2025, VLSBench, OD-WeaponDetection, Open Images V5, and an X-ray contraband collection derived from PIDray and SIXray (Sharifi, Zibaei, and Rezaei 2021; Ha et al. 2024; Omiotek 2025; Hu et al. 2025; Olmos, Tabik, and Herrera 2018; Kuznetsova et al. 2020; Manaenkov 2025;

**ImageNet-100**

| Method | LLaVA-1.5-7B Forget F-M↓ | F-Y↓ | F-O↓ | Capability C-M↑ | C-Y↑ | C-O↑ | General OBQA↑ | RWQA↑ | Qwen2.5-VL-7B Forget F-M↓ | F-Y↓ | F-O↓ | Capability C-M↑ | C-Y↑ | C-O↑ | General OBQA↑ | RWQA↑ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Vanilla | 98.00 | 91.20 | 16.40 | 96.50 | 90.10 | 52.80 | 66.00 | 52.50 | 99.50 | 92.10 | 22.40 | 98.20 | 93.50 | 66.80 | 85.50 | 67.50 |
| GA | 18.50 | 51.20 | 1.50 | 19.20 | 52.40 | 5.80 | 48.00 | 38.50 | 22.50 | 52.10 | 3.80 | 21.40 | 51.80 | 6.20 | 55.00 | 42.00 |
| GA-Diff | 16.80 | 50.80 | 1.20 | 17.50 | 51.60 | 4.90 | 47.50 | 37.00 | 19.80 | 51.50 | 3.20 | 18.90 | 50.90 | 5.40 | 54.00 | 40.50 |
| KL-Min | 65.00 | 55.00 | 1.50 | 88.50 | 80.20 | 40.50 | 63.00 | 48.00 | 62.00 | 58.00 | 2.50 | 15.20 | 45.00 | 0.00 | 78.00 | 25.00 |
| NPO | 64.00 | 54.50 | 1.20 | 90.20 | 82.10 | 42.80 | 64.00 | 49.50 | 52.00 | 55.00 | 1.50 | 92.50 | 86.20 | 48.50 | 82.00 | 55.00 |
| CLIPErase | 92.00 | 62.00 | 2.00 | 95.50 | 85.40 | 45.20 | 65.00 | 50.50 | 70.00 | 56.00 | 2.00 | 93.80 | 87.50 | 42.00 | 81.50 | 55.50 |
| MMU | 28.00 | 65.00 | 1.00 | 85.20 | 74.80 | 33.50 | 61.00 | 45.50 | 65.00 | 55.00 | 3.00 | 18.50 | 44.00 | 0.00 | 79.00 | 24.00 |
| CMU | 15.22 | 49.77 | 0.26 | 97.07 | 88.46 | 51.55 | 67.50 | 53.43 | 4.00 | 50.35 | 0.23 | 99.36 | 94.41 | 66.14 | 86.93 | 67.55 |

**RVC**

| Method | LLaVA-1.5-7B Forget F-M↓ | F-Y↓ | F-O↓ | Capability C-M↑ | C-Y↑ | C-O↑ | General OBQA↑ | RWQA↑ | Qwen2.5-VL-7B Forget F-M↓ | F-Y↓ | F-O↓ | Capability C-M↑ | C-Y↑ | C-O↑ | General OBQA↑ | RWQA↑ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Vanilla | 91.50 | 89.80 | 14.60 | 97.20 | 90.80 | 53.40 | 66.00 | 52.80 | 98.40 | 86.20 | 14.80 | 98.60 | 93.40 | 65.80 | 85.00 | 67.20 |
| GA | 15.20 | 52.40 | 2.10 | 16.80 | 51.90 | 4.50 | 48.50 | 39.00 | 18.50 | 53.20 | 2.80 | 19.40 | 52.50 | 5.60 | 52.00 | 41.00 |
| GA-Diff | 14.50 | 51.80 | 1.90 | 15.60 | 51.20 | 3.80 | 47.50 | 38.00 | 17.20 | 52.60 | 2.40 | 18.10 | 51.80 | 4.90 | 51.00 | 40.00 |
| KL-Min | 62.00 | 65.00 | 3.50 | 91.50 | 82.40 | 42.00 | 63.50 | 49.00 | 55.00 | 66.00 | 1.50 | 93.00 | 85.50 | 38.00 | 80.00 | 45.00 |
| NPO | 63.00 | 63.50 | 3.00 | 92.80 | 83.50 | 43.50 | 64.00 | 48.00 | 96.00 | 82.00 | 15.50 | 95.50 | 88.00 | 42.00 | 81.50 | 58.00 |
| CLIPErase | 91.00 | 87.00 | 16.50 | 96.00 | 86.50 | 46.00 | 65.00 | 50.50 | 68.00 | 60.00 | 4.50 | 95.50 | 89.00 | 42.50 | 81.00 | 52.00 |
| MMU | 88.00 | 82.00 | 16.00 | 96.50 | 85.80 | 45.50 | 64.50 | 50.00 | 58.00 | 56.00 | 10.00 | 94.00 | 86.50 | 39.00 | 80.50 | 50.50 |
| CMU | 8.64 | 51.72 | 0.07 | 94.61 | 84.13 | 45.78 | 67.50 | 50.47 | 4.17 | 50.32 | 0.47 | 99.60 | 94.80 | 65.39 | 86.50 | 62.71 |

Table 2: Main comparison on ImageNet-100 and RVC. F/C denote forget/capability scores; M/Y/O denote MCQ, positive-query yes/no confirmation accuracy, and Open-Gen.

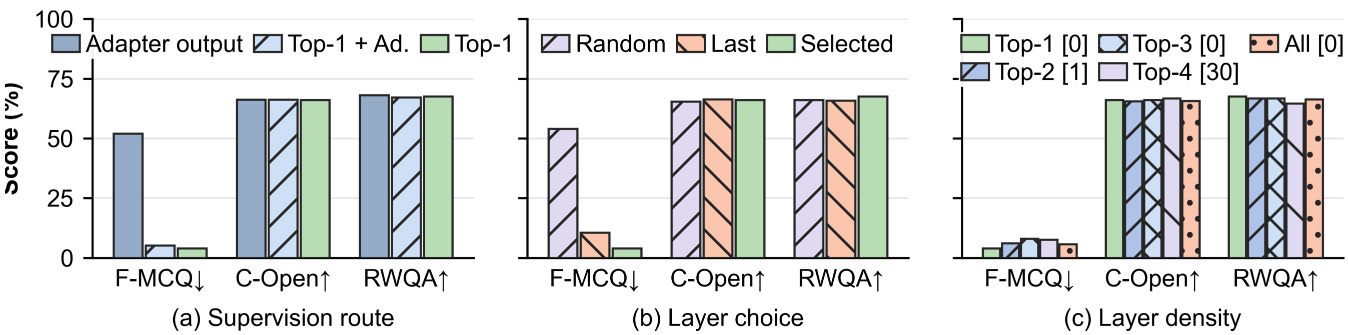



Figure 5: Matched layer-wise supervision ablation on Qwen2.5-VL-7B. **(a)** Adapter-output, selected Top-1, and Top-1-plus-adapter-output supervision. **(b)** Selected, random, and final-layer choices. **(c)** Increasing the density of semantic-layer supervision. Bars report class means; legend labels give invalid-output counts. Lower forget-side and higher capability-side values are preferred.

Wang et al. 2021; Miao et al. 2019). The forget source, retain-training source, and capability-evaluation sources remain separate throughout the pipeline. The complete domain-level composition and split counts, together with the taxonomy and preprocessing details, are reported in Appendix D.

All reported CMU results use one model-specific semantic layer: layer 22 for LLaVA-1.5-7B, layer 2 for Qwen2.5-VL-7B, and layer 35 for Qwen3-VL-8B. The adaptive anchor, retain, and parameter objectives are applied only at that layer; no loss is attached to the adapter output $H^{(0)}$. Each experiment retains its configuration and evaluation records.

## Baselines

We compare CMU with Gradient Ascent (GA) (Thudi et al. 2022; Jang et al. 2023), retain-regularized GA-Diff (Liu, Liu, and Stone 2022), KL-constrained forgetting (Maini et al. 2025), Negative Preference Optimization (NPO) (Zhang et al. 2024), MMU (Huo et al. 2025), and CLIPErase (Yang et al. 2025). Appendix E specifies the adapted trainable scopes, sample counts, batch sizes, and mask estimation.

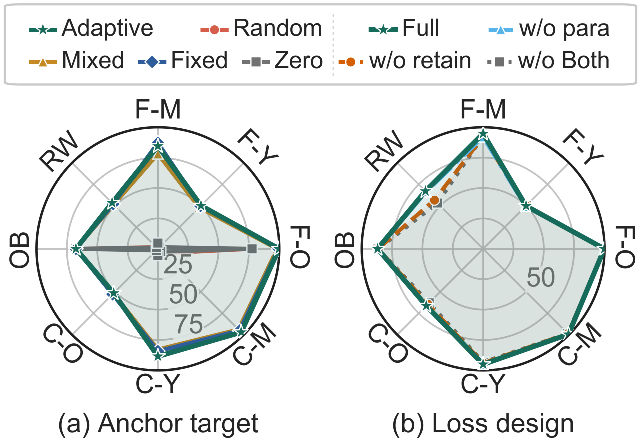



Figure 6: Matched diagnostics for **(a)** anchor targets and **(b)** loss components. Each axis shows a percentage-scale desirability score: $100 - s$ for the raw forget-side score $s$ and $s$ itself for capability and QA, without min-max normalization; larger radii are better.

## Metrics

We evaluate both target-class forgetting (F-) and held-out capability (C-) with MCQ, yes/no, and open-ended generation, together with OpenBookQA (Mihaylov et al. 2018) and RealWorldQA (xAI 2024). Each yes/no item asks whether the image belongs to its ground-truth class $C$, so the reference answer is always "Yes." F-Y and C-Y therefore measure positive-query class-confirmation accuracy rather than balanced binary accuracy: lower F-Y indicates reduced target-class confirmation, whereas higher C-Y indicates preserved non-target confirmation. Appendix A specifies exact parsing, invalid-output handling, ROUGE-L computation, and class aggregation. Lower F-MCQ and F-Open values also indicate better forgetting; higher capability-side and general QA values indicate better preservation.

## Main Results

**Comparison with baselines.** On ImageNet-100 and RVC, GA and GA-Diff achieve strong forgetting but severely degrade capability and general visual question answering. KL-Min and NPO preserve more capability but retain substantial target-class recognition. CLIPErase provides limited forgetting, while MMU is effective mainly on ImageNet-100 and transfers poorly to RVC. In contrast, CMU maintains a strong balance between forgetting and preservation across both benchmarks.

**Cross-benchmark and cross-model consistency.** CMU shows a consistent pattern of target reduction and capability retention on LLaVA and Qwen2.5 across both benchmarks. Appendix B reports full Qwen3-VL-8B results on both datasets and all eight metrics, exhibiting the same trend across data domains and model families.

**Prompt robustness.** CMU also remains robust to the tested prompt-only attacks, with only small changes in forget-side scores and stable capability. Appendix C reports every attack variant and its forget/capability scores.

## Ablation Studies

We conduct ablation studies on three key design choices: layer-wise supervision, anchor alignment, and the loss components. In each study, we hold all other variables fixed and change only the component under study.

**Ablation on Layer-wise Supervision.** We conduct three experiments to justify supervision at the selected Top-1 semantic layer. Since the adapter is the only updated component, directly supervising its output is a natural baseline. First, Figure 5(a) compares supervision at the selected layer, at the adapter output, and at both positions. Second, Figure 5(b) compares the selected layer with random and final-LLM-layer choices. Third, Figure 5(c) expands supervision from Top-1 to the Top-2, Top-3, Top-4, and all candidate layers.

The selected layer outperforms adapter-output supervision, while combining both positions brings no clear improvement. It also provides a better overall trade-off between forgetting and preservation than random or final-layer supervision. Supervising additional layers does not improve performance and introduces unnecessary computation and method complexity. These results support using only the selected Top-1 semantic layer in CMU.

**Ablation on Anchor Targets.** To assess adaptive anchor alignment, we compare it with fixed, mixed, zero, and random-unit targets under the same LLaVA training and evaluation protocol. Figure 6(a) shows that adaptive anchor alignment yields a comparatively balanced profile across the reported forget and capability measures. Fixed and mixed targets offer less balanced trade-offs, while zero and random-unit targets often produce invalid generations. These results support matching to a diverse neutral distribution instead of forcing every forget representation toward a single artificial target.

**Ablation on Loss Components.** We compare the full objective with variants that remove retain preservation, weight regularization, or both. Figure 6(b) shows that retain preservation supports capability, weight regularization supports forgetting, and removing both gives the weakest overall balance.

# Conclusion

We introduce CMU, an adapter-only method for reducing image-conditioned target-class recognition in adapter-based vision-language models. CMU restricts optimization to the lightweight vision-language adapter, selects one semantic LLM layer using class separability, and applies adaptive anchor alignment to match forget representations to a balanced neutral anchor distribution only at that layer. Retain-representation preservation and adapter regularization constrain representation and parameter drift. Across the evaluated settings, the reported results show a consistent pattern of target reduction and capability retention. Because the vision encoder and LLM remain frozen, the claim is limited to the image-conditioned adapter pathway rather than complete removal of target knowledge from the full model.

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